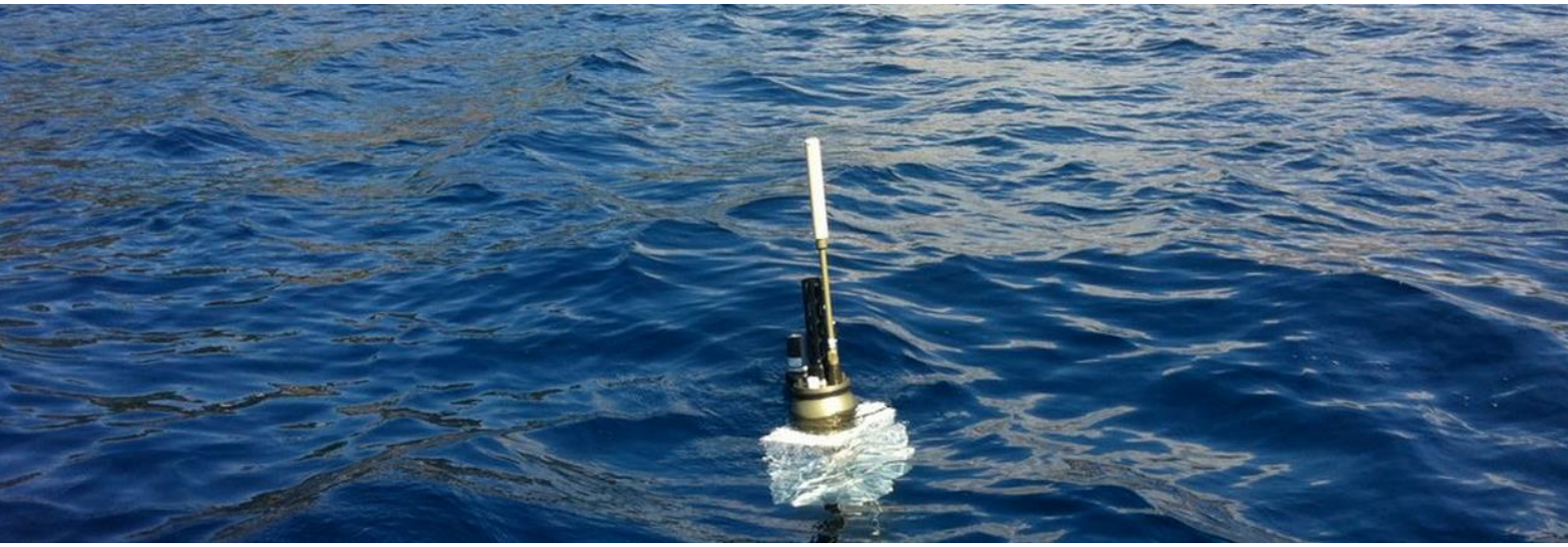
APMT Profiler – Event-Driven Capabilities

AUTOMATED **M**ULTI-**T**ASK **P**ROFILER



33-16-072_APMT_Event-Driven_Capabilities
Revision 1.0 (2024-05-15)

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1. Revision history

Revision	Release date	Notes	Author
1.0	2023-10-05	Original	C. SCHAEFFER

2. Introduction

The main objective of event-driven algorithm is to observe the impact on the environment of specific events (meteorological events, particle-flux event…):

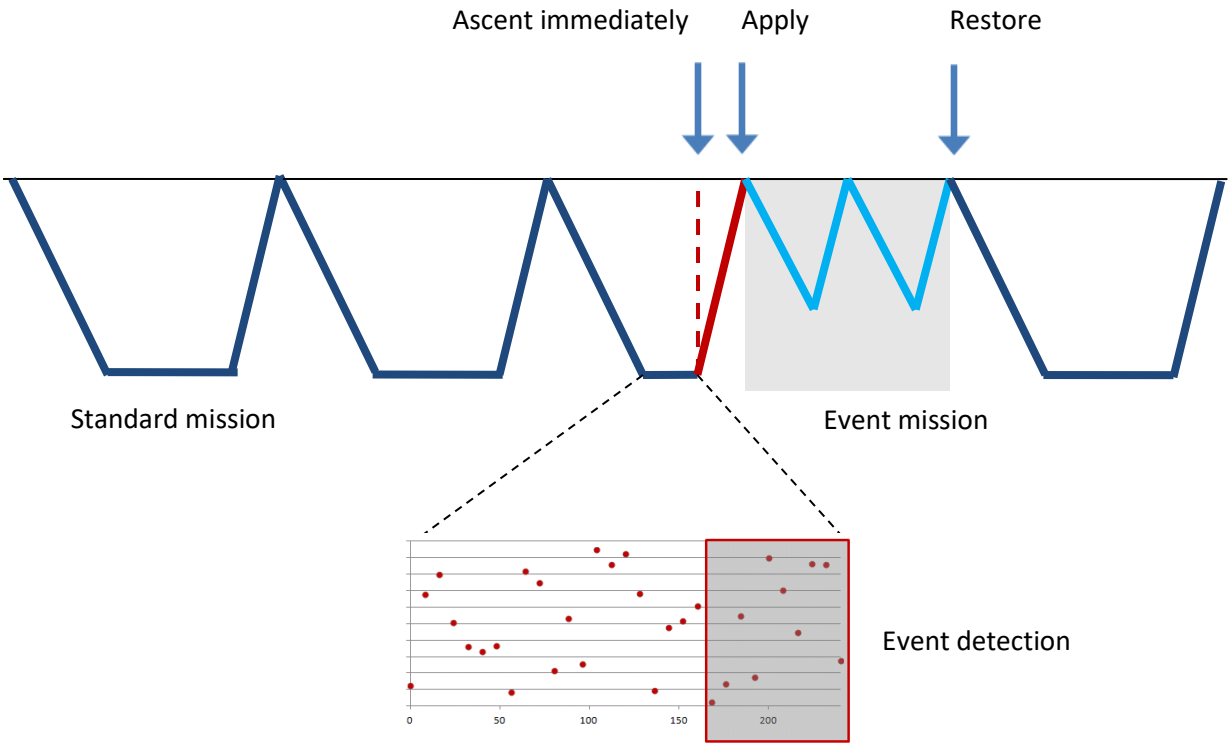
- Phytoplankton response
- Sea surface freshening
- Air-sea gas flux
- Vertical mixing

Events are detected while the float drifts at its parking depth.

3. Event-driven features

3.1 Overview

When an event occurs, the current profile is shortened, and the float immediately begins to ascend. After transmission, the specific settings associated to the event are applied and the event mission starts for a limited number of profiles. After completion of the event mission the standard mission settings are restored.

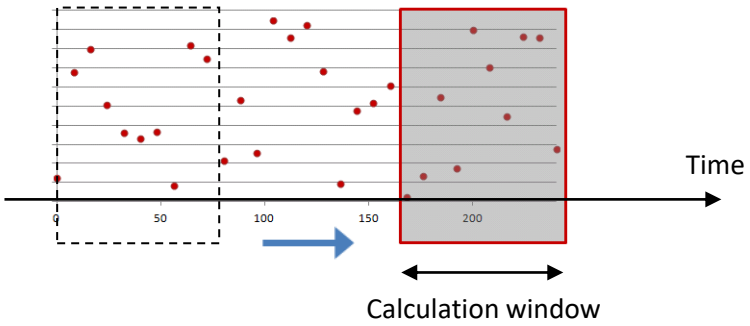


--- Figure 1. Event-driven overview ---

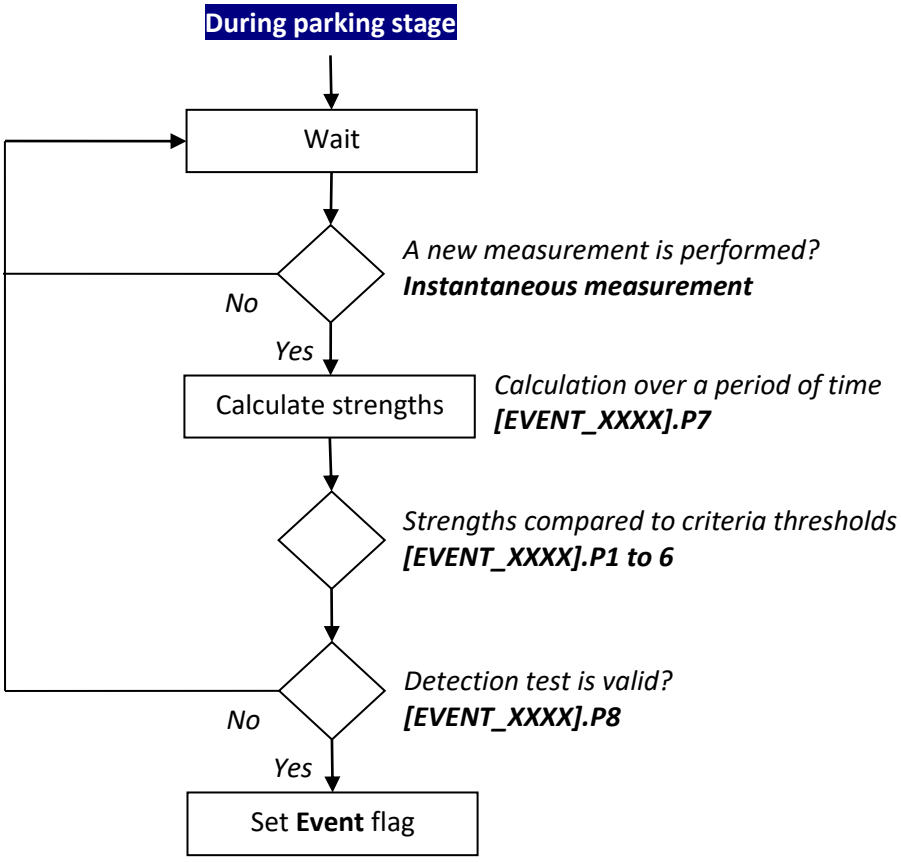
Comment: There is no new event detection while an event mission is in progress.

3.2 Event detection

A trigger event is defined as the occurrence of an event of a given strength during a given period of time (calculation window).



Strength is calculated during parking stage after each new acquisition sample of the sensor and the detection test is evaluated.



--- Figure 2. Event detection process ---

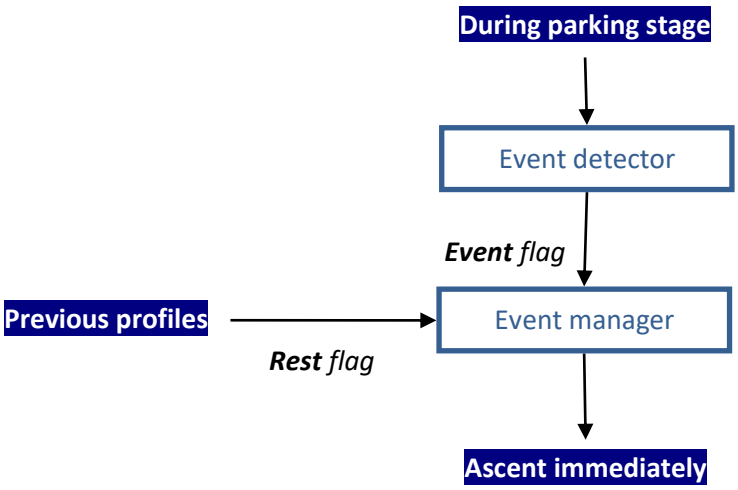
Several methods are available to detect events. The detection strategy is defined as a combination of criteria. Each criterion test is activated by a corresponding bit in a 16-bit configuration mask:

bit	Detection strategy [EVENT_XXXX].P8
0	The average strength is below the threshold over a given period
1	The median strength is below the threshold over a given period
2	The cumulative strength is below the threshold over a given period
3	The minimum strength is below the threshold over a given period
4	The maximum strength is below the threshold over a given period
5	The variability strength is below the threshold over a given period
6	The average strength exceeds the threshold over a given period
7	The median strength exceeds the threshold over a given period
8	The cumulative strength exceeds the threshold over a given period
9	The minimum strength exceeds the threshold over a given period
10	The maximum strength exceeds the threshold over a given period
11	The variability strength exceeds the threshold over a given period
12	The time since the last detection exceeds the calm period

Example:

- Low wind = 0x0001
- Storm = 0x0080
- Rain = 0x0100

After detection of an event, a rest period prevents the float from detecting successive events.



--- Figure 3. Event-driven decision process ---

3.3 Mission settings: apply and restore

The specific parameters to be applied in case of an event are stored in the event file. This file can contain parameters for multiple event sources (one of each):

_event.txt

```
1  !wait-event:rain
2  !param-pattern_01-1:500
3  !param-pattern_01-2:500
4  !param-pattern_01-3:86400
5  !param-sensor_04-16:00
6  !param-sensor_04-25:00
7  !wait-event:wind
8  !param-pattern_01-2:1000
9  !param-pattern_01-3:432000
```

--- Figure 4. Event file ---

The mechanism to restore the standard mission (after completion of the event mission) can be managed by the float itself (stand alone mode) or remotely (manual or server mode).

4. Parameter settings

4.1 Related parameters

[EVENT_DRIVEN]

No.	Comment	Limitation
0	Enable/disable event-driven feature	[True/False]
1	Event polling period (s) = maximum propagation delay between event detection and immediate ascent	[1-60] min

[EVENT_RAIN]

No.	Comment	Limitation
0	Enable/disable rain detection	[True/False]
1	Average criteria threshold	[0.0-9999.9] mm/h
2	Median criteria threshold	[0.0-9999.9] mm/h
3	Cumulative criteria threshold	[0.0-9999.9] mm/h
4	Minimum criteria threshold	[0.0-9999.9] mm/h
5	Maximum criteria threshold	[0.0-9999.9] mm/h
6	Variability criteria threshold	[0.0-9999.9] mm/h
7	Strength calculation period (s)	[1-96] hours
8	Criteria combination strategy	0xHHHH mask
9	Event mission duration (number of cycles)	[1-30]
10	Rest period (s)	[0-30] days
11	Activate/deactivate the stand-alone restore mode	[True/False]
12	Calm period (s)	[0-30] days

[EVENT_WIND]

No.	Comment	Limitation
0	Enable/disable wind detection	[True/False]
1	Average mean criteria threshold	[0.0-9999.9] m/s
2	Median criteria threshold	[0.0-9999.9] m/s
3	Cumulative criteria threshold	[0.0-9999.9] m/s
4	Minimum criteria threshold	[0.0-9999.9] m/s
5	Maximum criteria threshold	[0.0-9999.9] m/s
6	Variability criteria threshold	[0.0-9999.9] m/s
7	Strength calculation period (s)	[1-96] hours
8	Criteria combination strategy	0xHHHH mask
9	Event mission duration (number of cycles)	[1-30]
10	Rest period (s)	[0-30] days
11	Activate/deactivate the stand-alone restore mode	[True/False]
12	Calm period (s)	[0-30] days

Note: For more details on parameters refer to “33-16-048_Parameter_Set” manual.

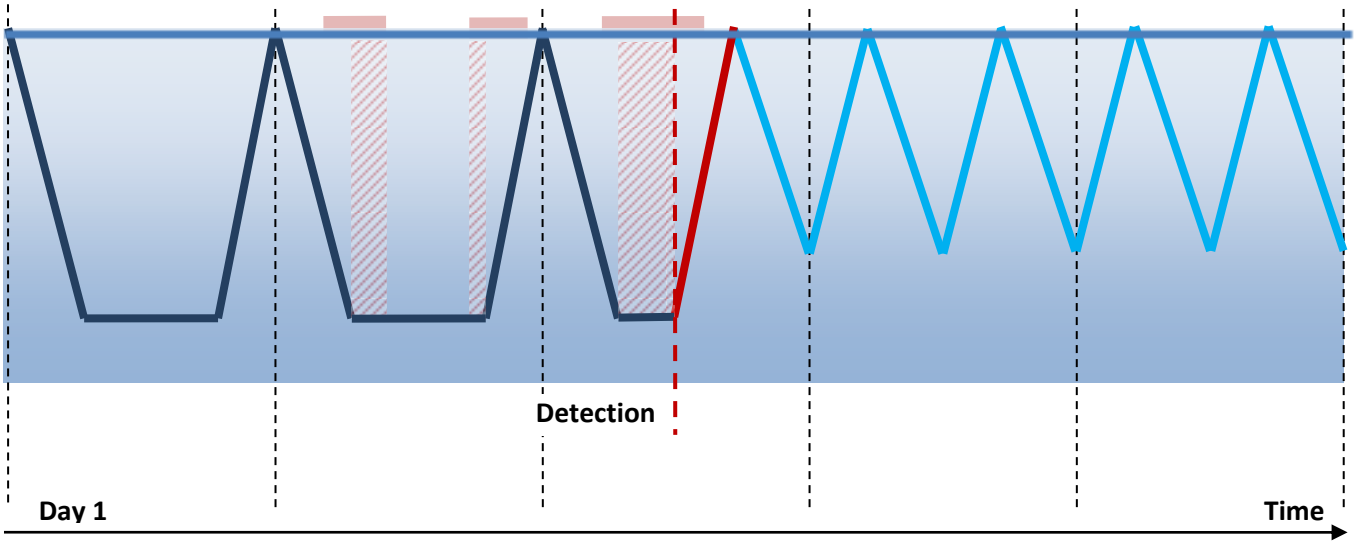
5. Appendices

The examples below illustrate event-driven mechanisms (specific 12-hour event mission) and the impact of different parameters applied to a daily profile mission:

- Represents the presence of an event at the surface
- Represents the standard mission
- Represents the event mission
- Represents an ascent stage following the detection of an event (no previous detection)
- Represents a parking stage following the detection of an event (while rest flag is pending)
- Represents a parking stage following the detection of an event (while calm period is not respected)

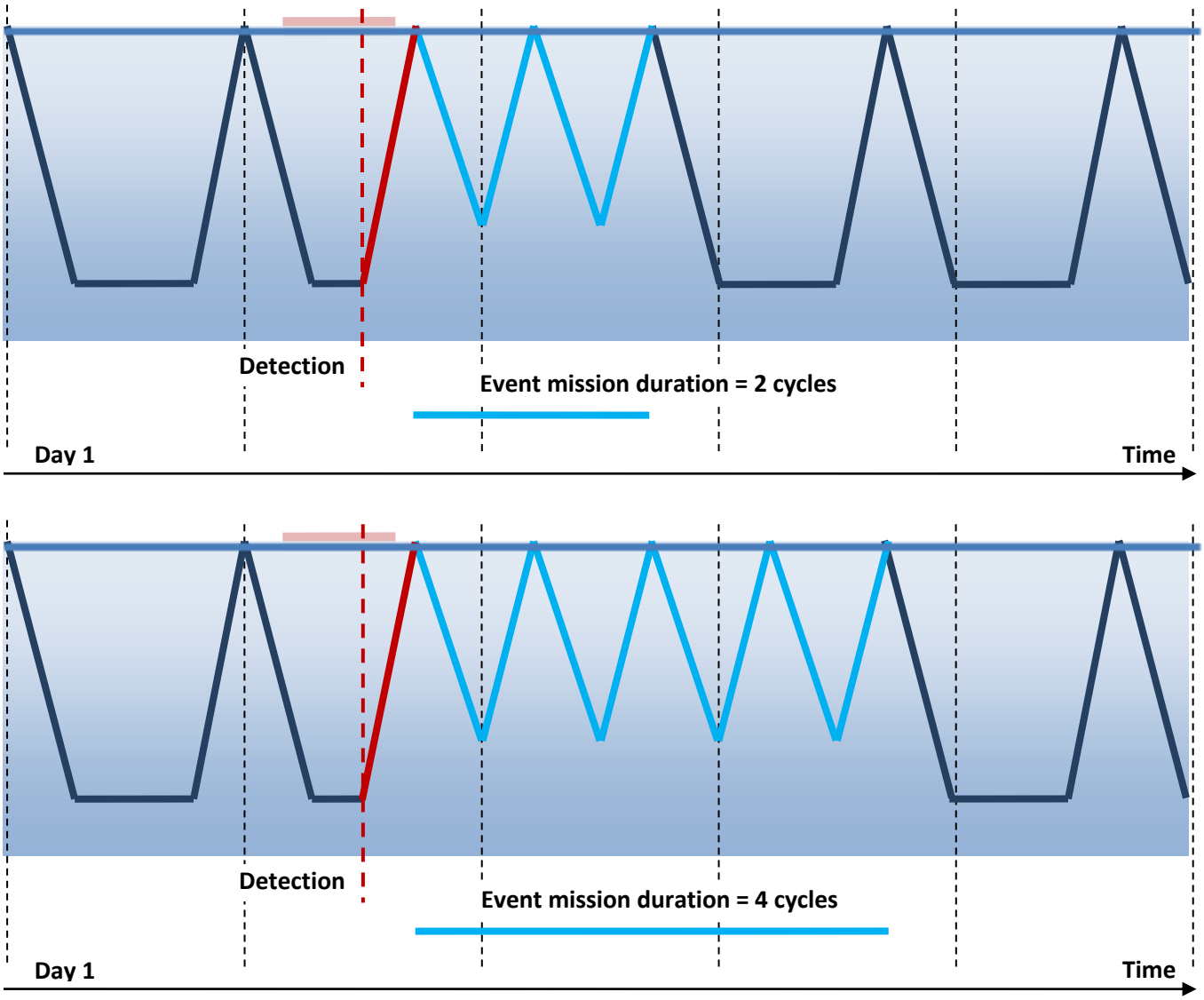
5.1 Example 1: [EVENT_XXXX].P1-8 parameters

These parameters are used to set the detection of an event of a sufficient level and duration.



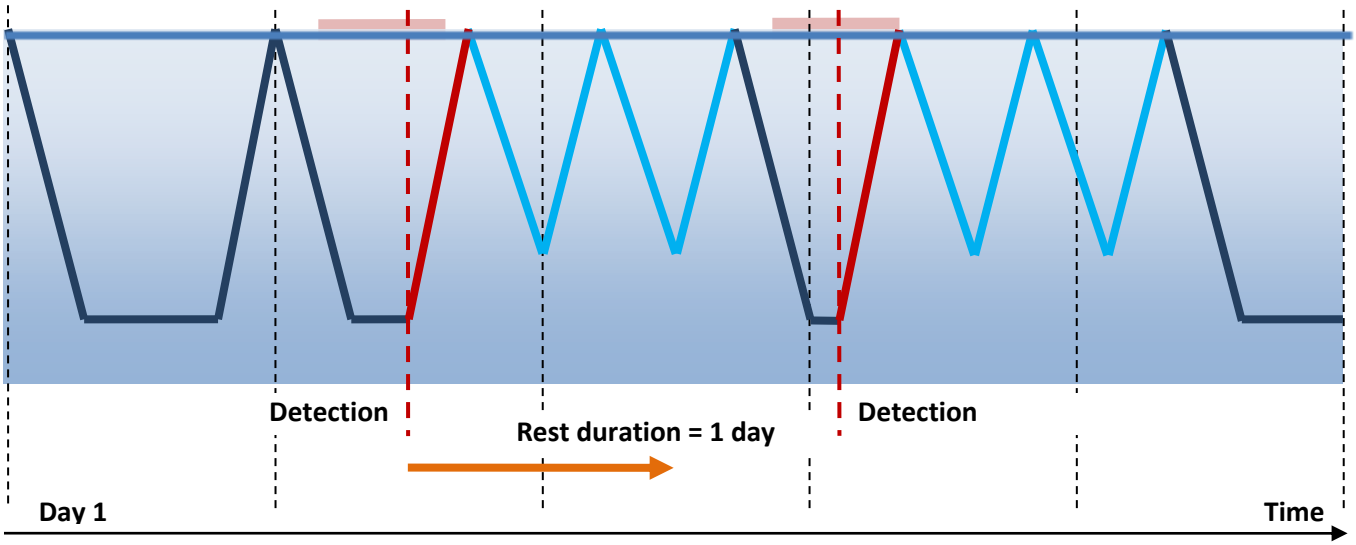
5.2 Example 2: [EVENT_XXXX].P9 parameter

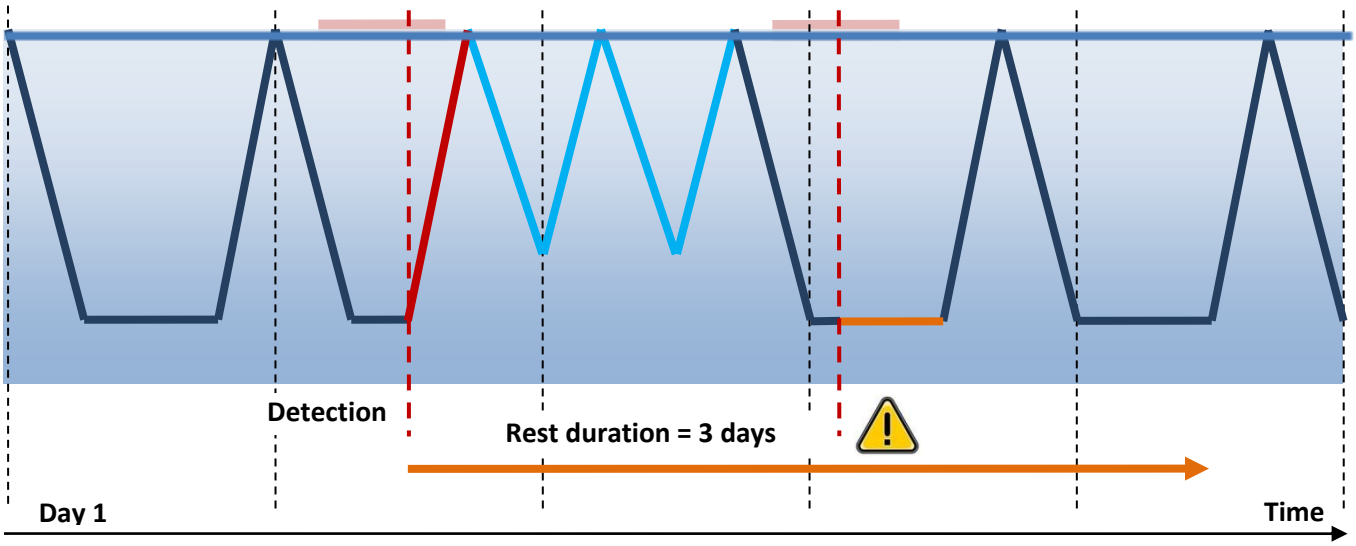
This parameter is used to select the duration of the event mission.



5.3 Example 3: [EVENT_XXXX].P10 parameter

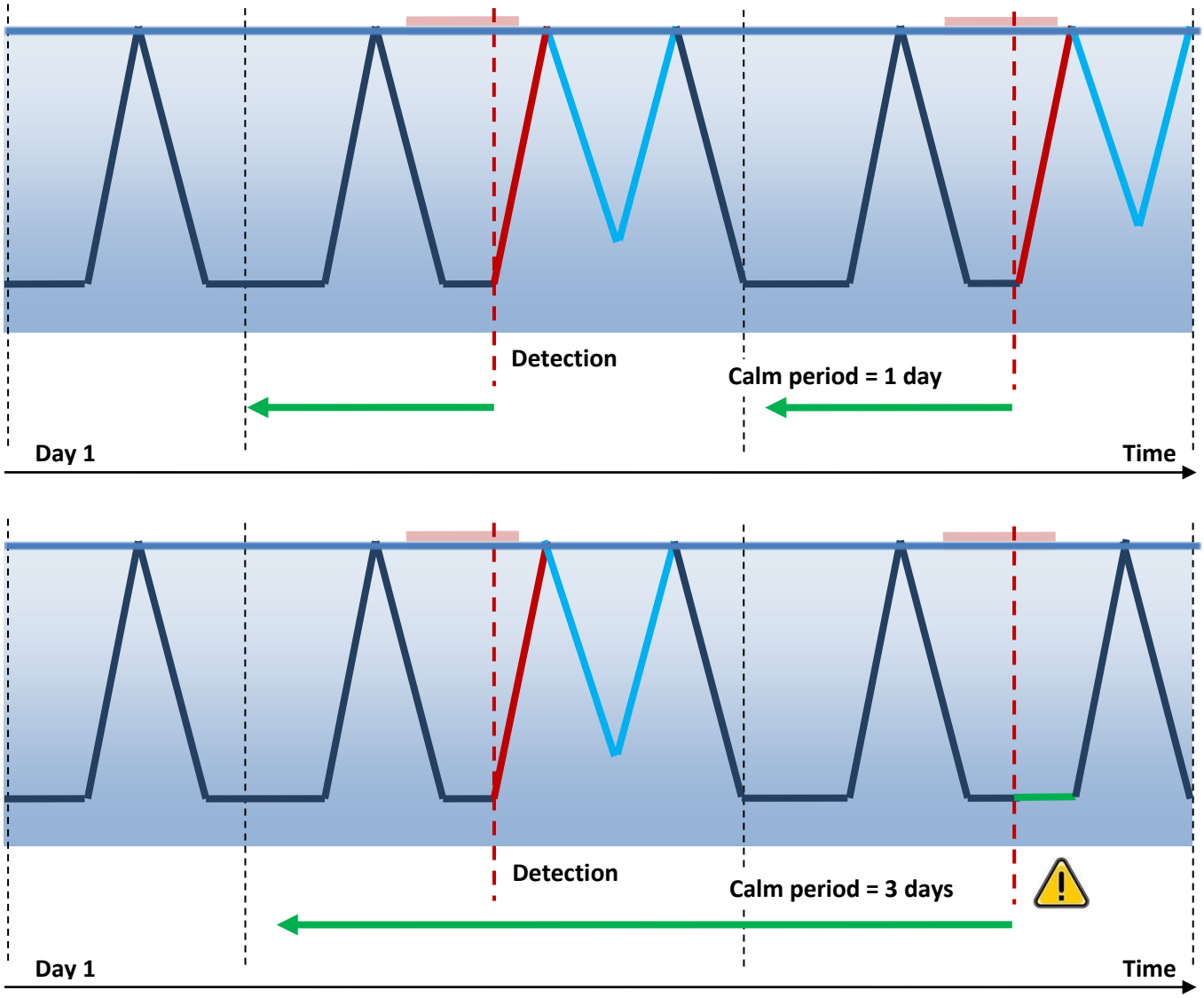
This parameter is used to hide the detection of a new event following a previous detection.





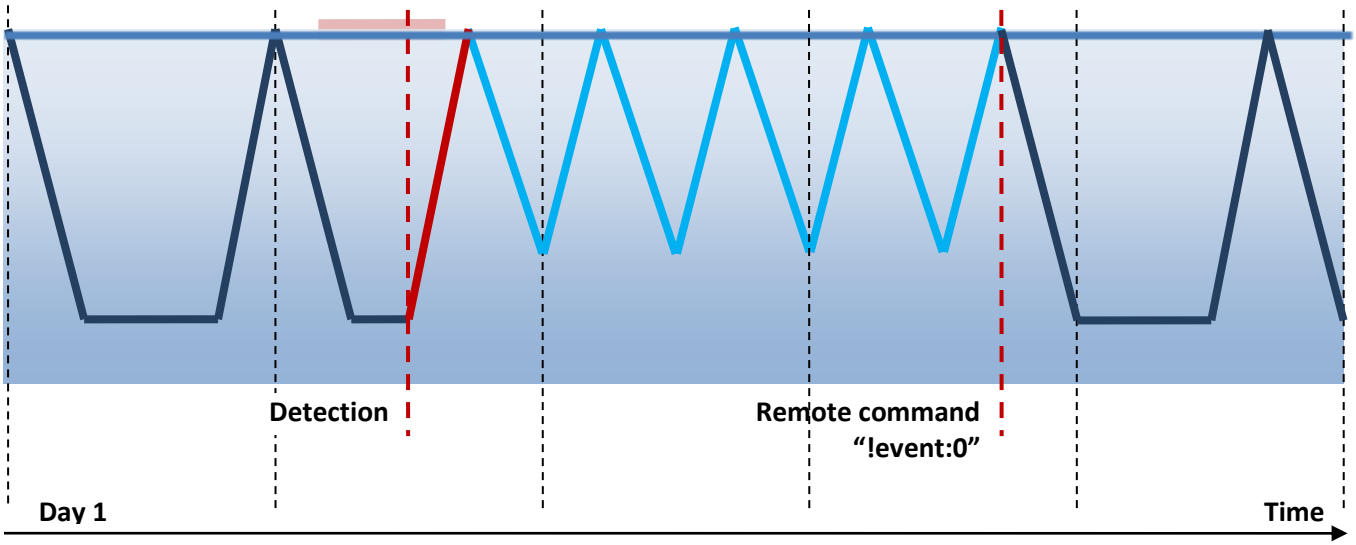
5.4 Example 4: [EVENT_XXXX].P12 parameter

This parameter is used to enable the detection of an event only after a period of calm (no previous detection).



5.5 Example 5: [EVENT_XXXX].P11 parameter in “server mode”

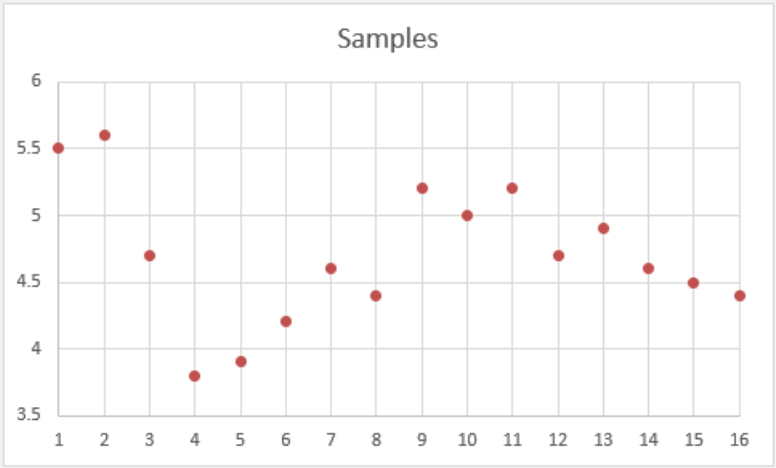
This parameter is used to select the stand-alone or manual/server mode.



5.6 Calculation methods

Here is an illustration of the results of the calculation methods applied to a set of 16 samples.

Samples	Calculation method	Result
5.5	Average	4.70
5.6	Median	4.65
4.7	Sum	75.2
3.8	Minimum	3.8
3.9	Maximum	5.6
4.2	Standard Deviation	0.518
4.6		
4.4		
5.2		
5.0		
5.2		
4.7		
4.9		
4.6		
4.5		
4.4		



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